Architecture Decision Records

MediCare — Phase 5 Technical Decisions, Risks & Open Questions

API-FIRST ARCHITECTURE | ASP.NET CORE | MICROSOFT AZURE

Document Control

Project	MediCare — Clinic Management & Appointment System
Phase	Phase 5 — Deployment & Integration
Architecture	API-First N-Tier Web Application
Version	1.0 — Proposed baseline

How to Read This Document

Items marked CONFIRMED are explicit project decisions or requirements. Items marked PROPOSED are recommended technical choices that require approval or implementation. Items marked OPTIONAL depend on scope and schema approval. Items marked OPEN require a team decision before the related implementation is treated as final.

1. Decision Status Legend

Status	Meaning
CONFIRMED	Explicit in user-confirmed architecture or project requirements.
PROPOSED	Recommended design; requires approval or implementation.
OPEN	Cannot be finalized from available sources.
OPTIONAL	Only if the related Should/Could feature is included.

ADR-001 — API-first separation

Field	Decision record
Status	CONFIRMED
Decision	MediCare.API is the complete backend; MediCare.Web is a separate MVC client.
Alternatives considered	MVC with direct EF Core was considered but rejected because user confirmed API-first architecture.
Rationale / trade-offs	Clear boundaries, reusable API, independent deployments. Adds HTTP/auth/token management complexity.
Consequences / follow-up	Keep all business rules and database access in API/Services/Data. Test MVC-to-API integration.

ADR-002 — N-Tier backend layering

Field	Decision record
Status	CONFIRMED
Decision	MediCare.API → MediCare.Services → MediCare.Data.
Alternatives considered	Microservices and direct controller-to-database access were rejected as unnecessary.
Rationale / trade-offs	Maintainable and appropriate for project size. Requires discipline to keep dependency direction.
Consequences / follow-up	API owns composition root; Services business logic; Data persistence.

ADR-003 — .NET target

Field	Decision record
Status	PROPOSED
Decision	Target .NET 10 LTS; use latest 10.0.x servicing patch available at build time.
Alternatives considered	Using .NET 11 RC is not recommended for production baseline; .NET 9 reaches end of support 10 Nov 2026.
Rationale / trade-offs	Longer support horizon and alignment with EF Core 10.
Consequences / follow-up	Verify hosting and package compatibility. Source code was not supplied; do not silently retarget existing projects.

ADR-004 — ORM and database

Field	Decision record
Status	PROPOSED
Decision	EF Core 10 + Azure SQL Database; local SQL Server 2022 Developer or a compatible supported SQL Server for development.
Alternatives considered	SQL Server-only self-hosted production was considered but Azure SQL is the documented cloud target.
Rationale / trade-offs	Managed backups and Azure integration. Service tier and costs must be chosen.
Consequences / follow-up	Preserve 11-table baseline unless explicit schema extensions are approved.

ADR-005 — Authentication topology

Field	Decision record
Status	PROPOSED
Decision	ASP.NET Core Identity owned by API; MVC secure cookie session; API JWT Bearer with short-lived access token and protected server-side refresh credential.
Alternatives considered	MVC local Identity and browser localStorage tokens rejected.
Rationale / trade-offs	API remains authority and tokens stay server-side for MVC calls. Requires secure refresh/session lifecycle.
Consequences / follow-up	Confirm actual Identity implementation, token storage, revocation, CSRF and cookie policies.

ADR-006 — Azure hosting

Field	Decision record
Status	PROPOSED
Decision	Separate App Service resources for Web and API; share an App Service Plan within each environment only if resource isolation and capacity are acceptable.
Alternatives considered	One monolithic Web/API deployable rejected. Separate plans per app/environment may increase cost.
Rationale / trade-offs	Independent deployment while controlling cost.
Consequences / follow-up	Staging and Production must have separate app settings, database and secrets.

ADR-007 — Real-time notifications

Field	Decision record
Status	CONFIRMED/PROPOSED
Decision	SignalR hub is hosted by API; Azure SignalR Service in Default mode is proposed as proxy.
Alternatives considered	Serverless mode is not suitable for an ASP.NET Core-hosted hub.
Rationale / trade-offs	Managed connection scale and backend-owned authorization.
Consequences / follow-up	Use trusted negotiate flow and short-lived hub access token; confirm actual connection topology.

ADR-008 — Background jobs

Field	Decision record
Status	PROPOSED
Decision	API-hosted BackgroundService for Pending expiry and reminders for initial release, with durable idempotent claims/leases.
Alternatives considered	In-memory timer without durable coordination rejected. Separate scheduler service deferred.
Rationale / trade-offs	Simple with current API architecture. Multi-instance processing needs coordination and Always On.
Consequences / follow-up	Validate worker execution guarantees and prevent duplicate notification/expiry actions.

ADR-009 — Appointment concurrency

Field	Decision record
Status	OPEN/PROPOSED
Decision	Use RowVersion for updates plus transaction/locking or slot reservation constraint that blocks overlapping interval inserts.
Alternatives considered	RowVersion alone is insufficient for two new rows that overlap.
Rationale / trade-offs	Protects business invariant at actual concurrent load.
Consequences / follow-up	Choose and test exact SQL mechanism against interval overlap, not only identical start-time uniqueness.

ADR-010 — Medical document storage

Field	Decision record
Status	OPTIONAL
Decision	Private Azure Blob Storage plus metadata table only if upload/download scope is implemented.
Alternatives considered	Public blob or database binary storage not selected as default.
Rationale / trade-offs	Secure, scalable file storage. Adds metadata, malware/content validation and retention work.
Consequences / follow-up	Approve MedicalDocuments extension and download authorization before enabling.

ADR-011 — Audit logging

Field	Decision record
Status	OPEN/PROPOSED
Decision	Persist dedicated audit events for each medical record read/change; AuditLogs schema extension or approved immutable audit store.
Alternatives considered	Ordinary Application Insights telemetry alone is not a business audit trail.
Rationale / trade-offs	Supports privacy traceability and incident investigation.
Consequences / follow-up	Define actor, action, resource, UTC time, outcome, correlation ID and retention; avoid PHI content.

ADR-012 — Notification reliability

Field	Decision record
Status	OPEN/PROPOSED
Decision	Persist delivery state/retries through approved Notification extension or outbox table; SignalR is best-effort and REST remains source of truth.
Alternatives considered	Direct fire-and-forget calls without retry state rejected.
Rationale / trade-offs	Resilient SMS/email retries. Requires schema choice and idempotency.
Consequences / follow-up	Baseline Notifications lacks Channel/DeliveryStatus/RetryCount/NextAttemptAt.

ADR-013 — API contract

Field	Decision record
Status	PROPOSED
Decision	Versioned REST API under /api/v1; OpenAPI 3.0.3 and Postman collection generated from contract.
Alternatives considered	Unversioned route set rejected.
Rationale / trade-offs	Clear consumer contract for MVC.
Consequences / follow-up	All listed paths are proposed until confirmed against controllers.

ADR-014 — CI/CD

Field	Decision record
Status	PROPOSED
Decision	GitHub Actions with separate Web/API artifacts, staging gate and production approval.
Alternatives considered	Single manual deployment step rejected.
Rationale / trade-offs	Fits common Git-based team workflow.
Consequences / follow-up	Confirm repository host. If Azure Repos, use Azure Pipelines.

ADR-015 — Reporting revenue

Field	Decision record
Status	OPEN
Decision	Report fee-based appointment aggregates only; do not call it collected cash revenue without payment ledger.
Alternatives considered	Payment subsystem is out of scope.
Rationale / trade-offs	Preserves documented reports without inventing payments.
Consequences / follow-up	Clarify if dashboard means estimated consultation fees or actually collected revenue.

ADR-016 — Time zone

Field	Decision record
Status	CONFIRMED
Decision	Persist UTC; display Cairo local time.
Alternatives considered	Store local wall-clock timestamps rejected.
Rationale / trade-offs	Unambiguous persistence and comparison.
Consequences / follow-up	Use IANA/Windows mapping correctly; test day-boundary conversions.